Failure Analysis Report

FAIL-0090 — Network Failure

1. Failure Summary

Failure ID	FAIL-0090
Failure Type	Network Failure
Severity	Trivial
Component	Sync Engine
Application	APP-005 — WearableHealth
Build Version	4.5.7
Timestamp	2024-10-24 10:09:47
Environment	ENV-003 — Hub Lab
OS Version	Tizen 7.0
Network	WiFi-6
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-0247
Title	Ensure session token refresh synchronizes correctly across devices when the targ...
Execution ID	EXEC-00397
Test Run	RUN-011 — Regression Cycle 11 — Build 6.3.9
Requirement	REQ-0230 — Network Behavior — Battery Optimization

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete NFC tap-to-pay within the defined SLA.
Observed Behaviour	The system exhibited network failure when two concurrent requests are made simultaneously.

4. Error Details

Error Message:

```
SocketException: Connection reset by peer on interface wlan0.
```

Stack Trace / Log Excerpt:

```
com.samsung.wallet.RenderingEngine.execute(line 675)
```

Log Reference	LOG-40134

5. Executed Steps at Time of Failure

SYNTHETIC DATA — No real Samsung confidential information

1	Simulate high CPU load using the network emulator profile '3G-Low...	The performance counter logs a value below the SLA threshold.
2	Verify that the UI displays 'Payment approved'.	A success toast message is displayed.
3	Disconnect the headset and wait 10 seconds.	The device reconnects automatically within 10 seconds.
4	Navigate to Settings > Bluetooth and tap 'Enable Sync'.	The operation completes without error.
5	Observe the status indicator for up to 13 seconds.	No crash or ANR is observed.
6	Simulate Bluetooth interference using the network emulator profil...	A success toast message is displayed.

6. Root Cause Analysis

Database migration v2.4 incorrectly updates the index on user_sessions, causing full table scans.

7. Linked Defect

Defect ID	BUG-0172
Title	Network Failure in Sync Engine: SocketException: Connection reset by peer on int...
Severity	Trivial
Priority	Critical
Status	OPEN
Root Cause	Database migration v2.4 incorrectly updates the index on user_sessions, causing ...
Resolution	—
Assigned To	ENG-097

8. Historical Similar Failures

HIST-0089	FAIL-0090	Network Failure	4.3.0	2023-09-29	15	Wont Fix
HIST-0170	FAIL-0171	Network Failure	6.1.4	2023-12-26	1	Resolved
HIST-0442	FAIL-0443	Network Failure	4.1.1	2024-04-17	13	Wont Fix

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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